

APP DEVELOPMENT

Code Snippets

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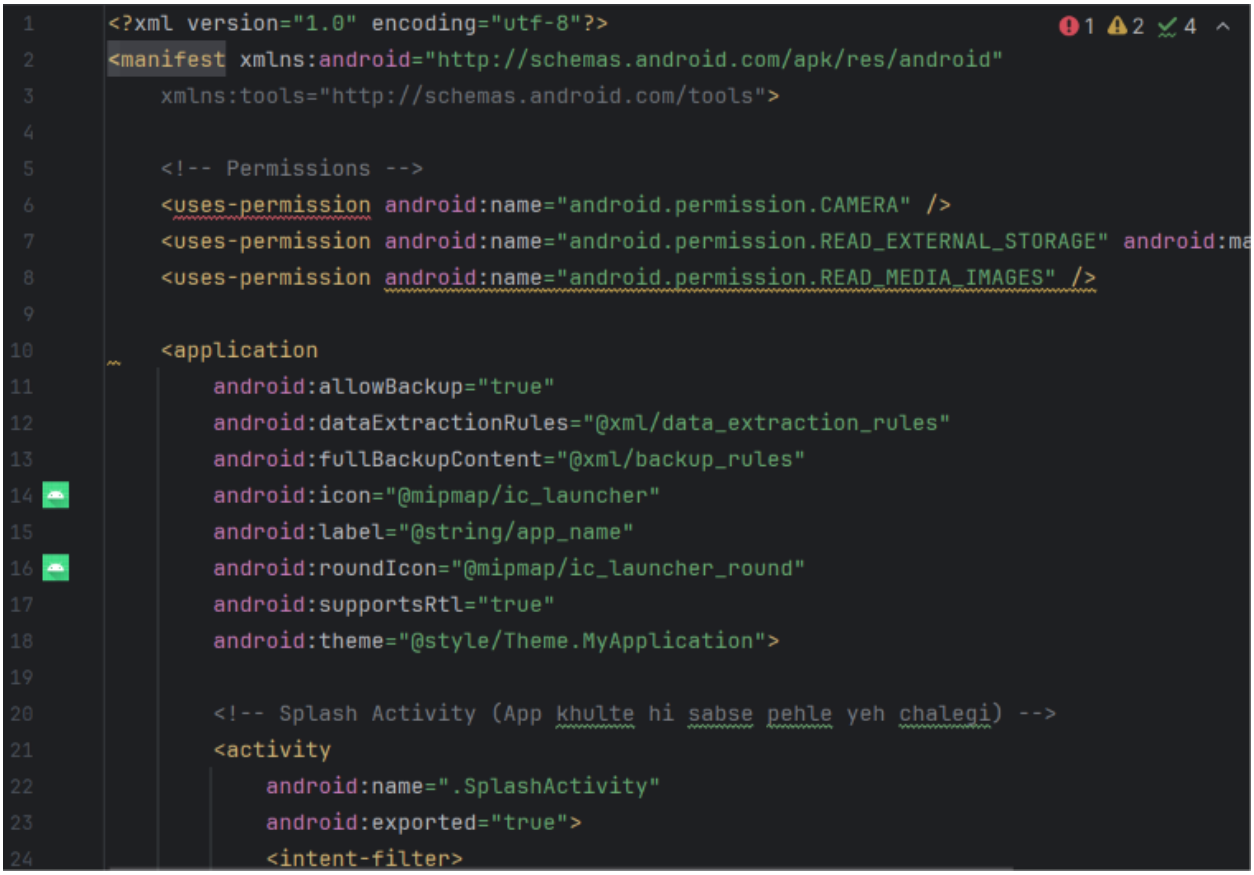
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Code Snippets

1. Android Manifest Configuration:

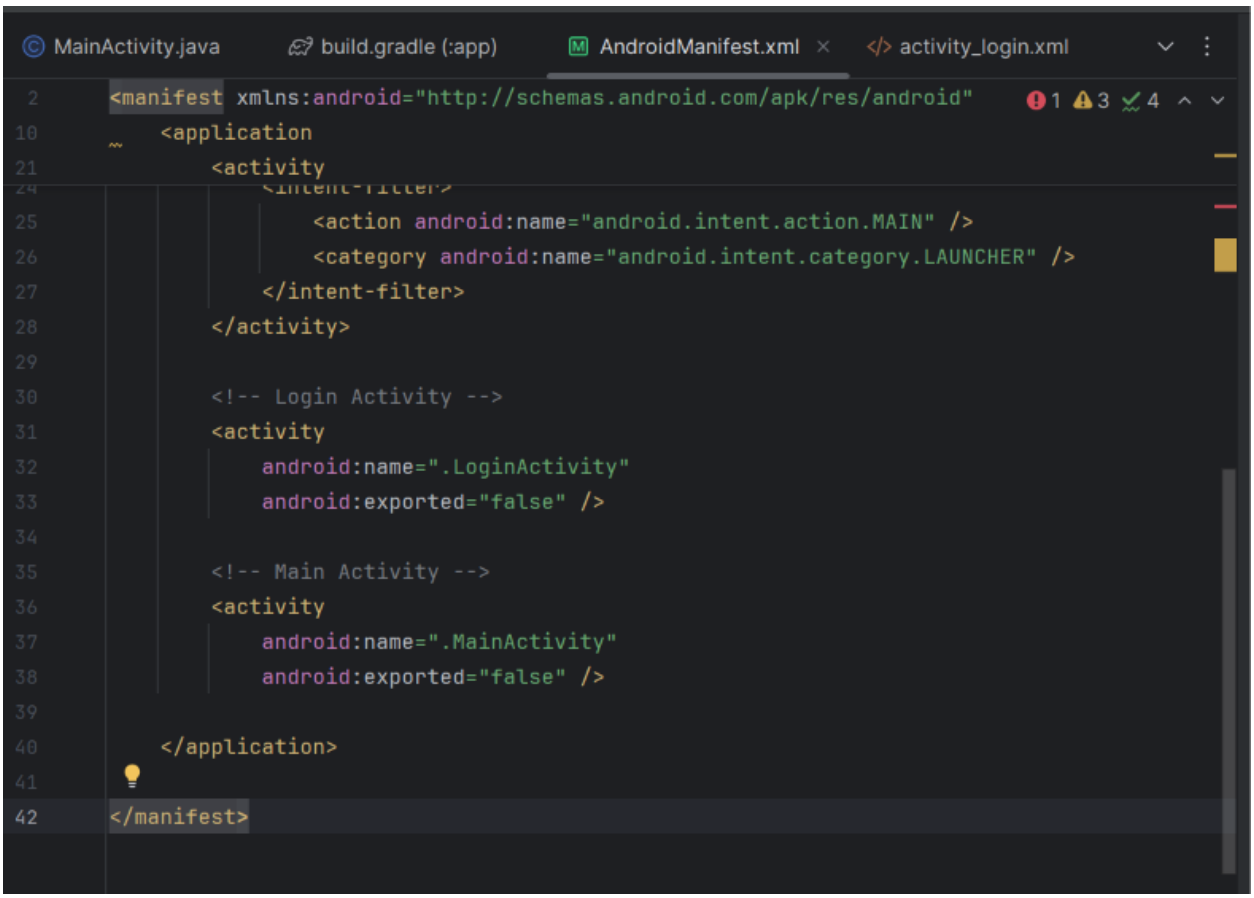


```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

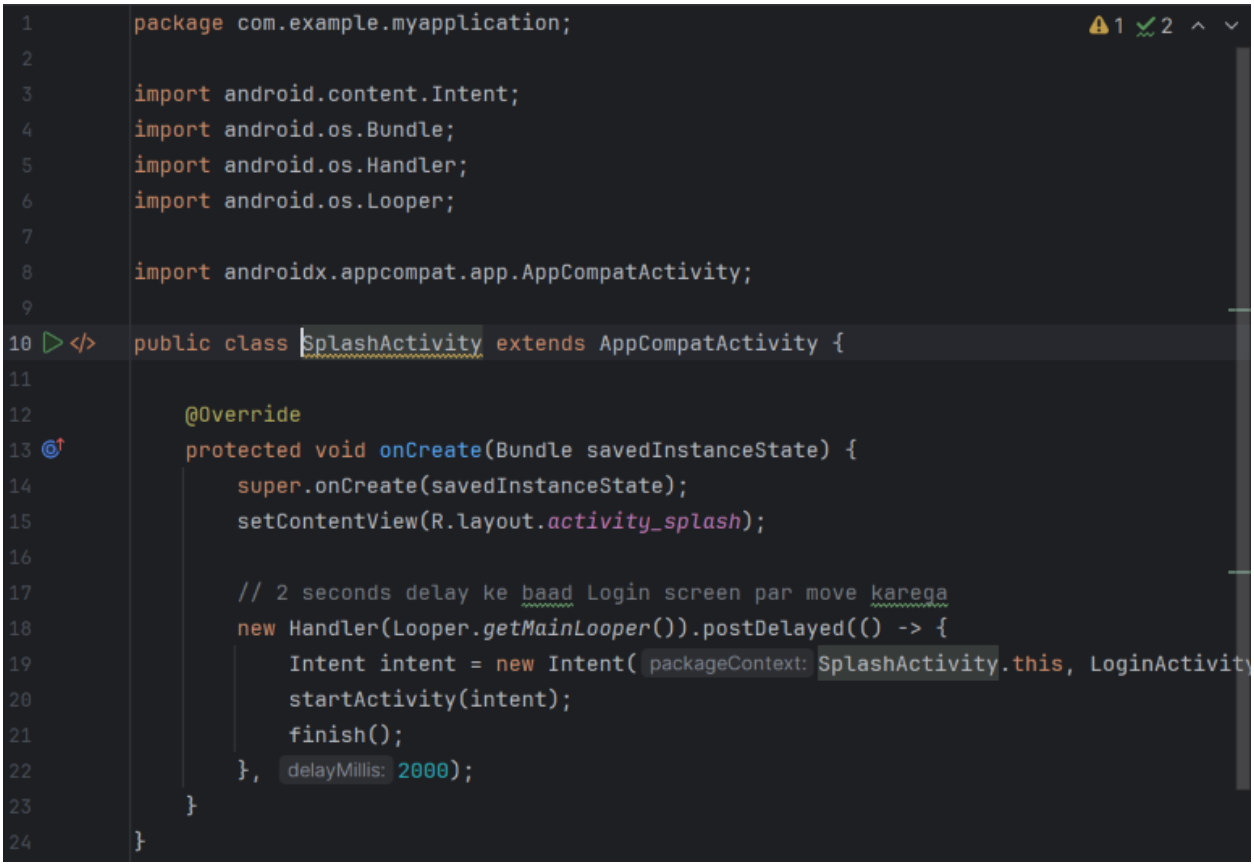
    <!-- Permissions -->
    <uses-permission android:name="android.permission.CAMERA" />
    <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" android:ma
    <uses-permission android:name="android.permission.READ_MEDIA_IMAGES" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.MyApplication">

        <!-- Splash Activity (App khulte hi sabse pehle yeh chalegi) -->
        <activity
            android:name=".SplashActivity"
            android:exported="true">
            <intent-filter>
```



2. Splash screen configuration



```java
package com.example.myapplication;

import android.content.Intent;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;

import androidx.appcompat.app.AppCompatActivity;

public class SplashActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_splash);

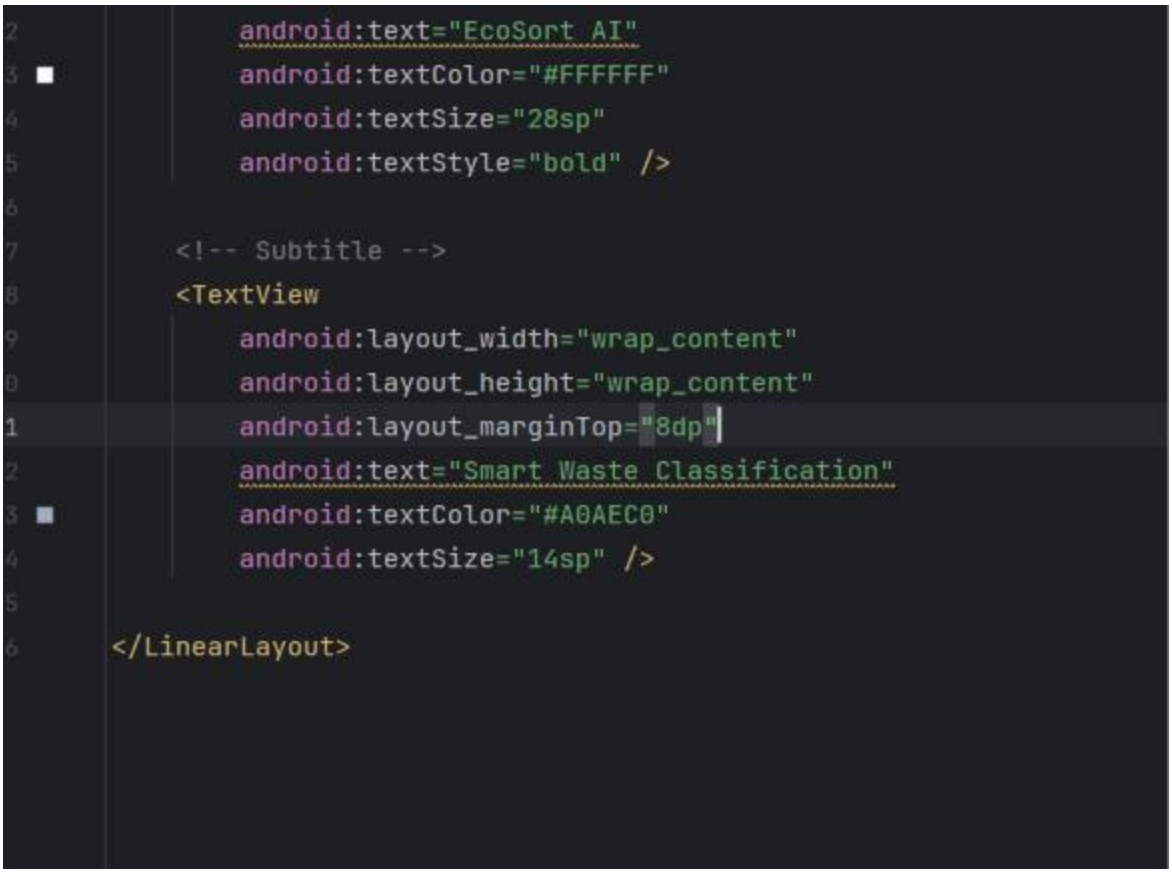
        // 2 seconds delay ke baad Login screen par move karega
        new Handler(Looper.getMainLooper()).postDelayed(() -> {
            Intent intent = new Intent( packageContext: SplashActivity.this, LoginActivity
            startActivity(intent);
            finish();
        }, delayMillis: 2000);
    }
}
```

Splash screen xml code

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#101426"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="24dp">

    <!-- App Logo / Icon -->
    <ImageView
        android:layout_width="120dp"
        android:layout_height="120dp"
        android:src="@drawable/ic_launcher_foreground"
        android:contentDescription="App Logo" />

    <!-- App Title -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="EcoSort AI"
        android:textColor="#FFFFFF"
        android:textSize="28sp"
        android:textStyle="bold" />
```



MainActivity configuration:

```java
package com.example.myapplication;

import android.content.Intent;
import android.content.res.AssetFileDescriptor;
import android.graphics.Bitmap;
import android.graphics.ImageDecoder;
import android.net.Uri;
import android.os.Build;
import android.os.Bundle;
import android.provider.MediaStore;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import org.tensorflow.lite.Interpreter;

```

```
import androidx.core.view.WindowInsetsCompat;                    4

import org.tensorflow.lite.Interpreter;

import java.io.BufferedReader;
import java.io.FileInputStream;
import java.io.IOException;
import java.io.InputStreamReader;
import java.nio.ByteBuffer;
import java.nio.ByteOrder;
import java.nio.MappedByteBuffer;
import java.nio.channels.FileChannel;
import java.util.ArrayList;
import java.util.List;

public class MainActivity extends AppCompatActivity {

    3 usages
    private ImageView previewImage;
    2 usages
    private Button selectImageButton, captureImageButton;
    5 usages
    private TextView resultText;
```

```
    2 usages
43      private ActivityResultLauncher<String> galleryLauncher;
    2 usages
44      private ActivityResultLauncher<Intent> cameraLauncher;
45
    3 usages
46      private Interpreter tfliteInterpreter;
    5 usages
47      private List<String> labelList;
    8 usages
48      private final int IMAGE_SIZE = 224;
49
50      @Override
51      protected void onCreate(Bundle savedInstanceState) {
52          super.onCreate(savedInstanceState);
53          EdgeToEdge.enable( $this$enableEdgeToEdge: this);
54          setContentView(R.layout.activity_main);
55
56          ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), ( View v,
57              Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars()
58              v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBar
59              return insets;
```

```
140     }
141
    2 usages
142     private void runClassification(Bitmap bitmap) {
143         if (tfliteInterpreter == null || labelList == null) {
144             resultText.setText("Model not initialized!");
145             return;
146         }
147
148         Bitmap resizedBitmap = Bitmap.createScaledBitmap(bitmap, IMAGE_SIZE, IMAGE_SIZE,
149         ByteBuffer byteBuffer = convertBitmapToByteBuffer(resizedBitmap);
150
151         float[][] outputMatrix = new float[1][labelList.size()];
152         tfliteInterpreter.run(byteBuffer, outputMatrix);
153
154         int maxIndex = 0;
155         float maxProbability = outputMatrix[0][0];
156         for (int i = 1; i < labelList.size(); i++) {
157             if (outputMatrix[0][i] > maxProbability) {
158                 maxProbability = outputMatrix[0][i];
159                 maxIndex = i;
160             }
161         }
```

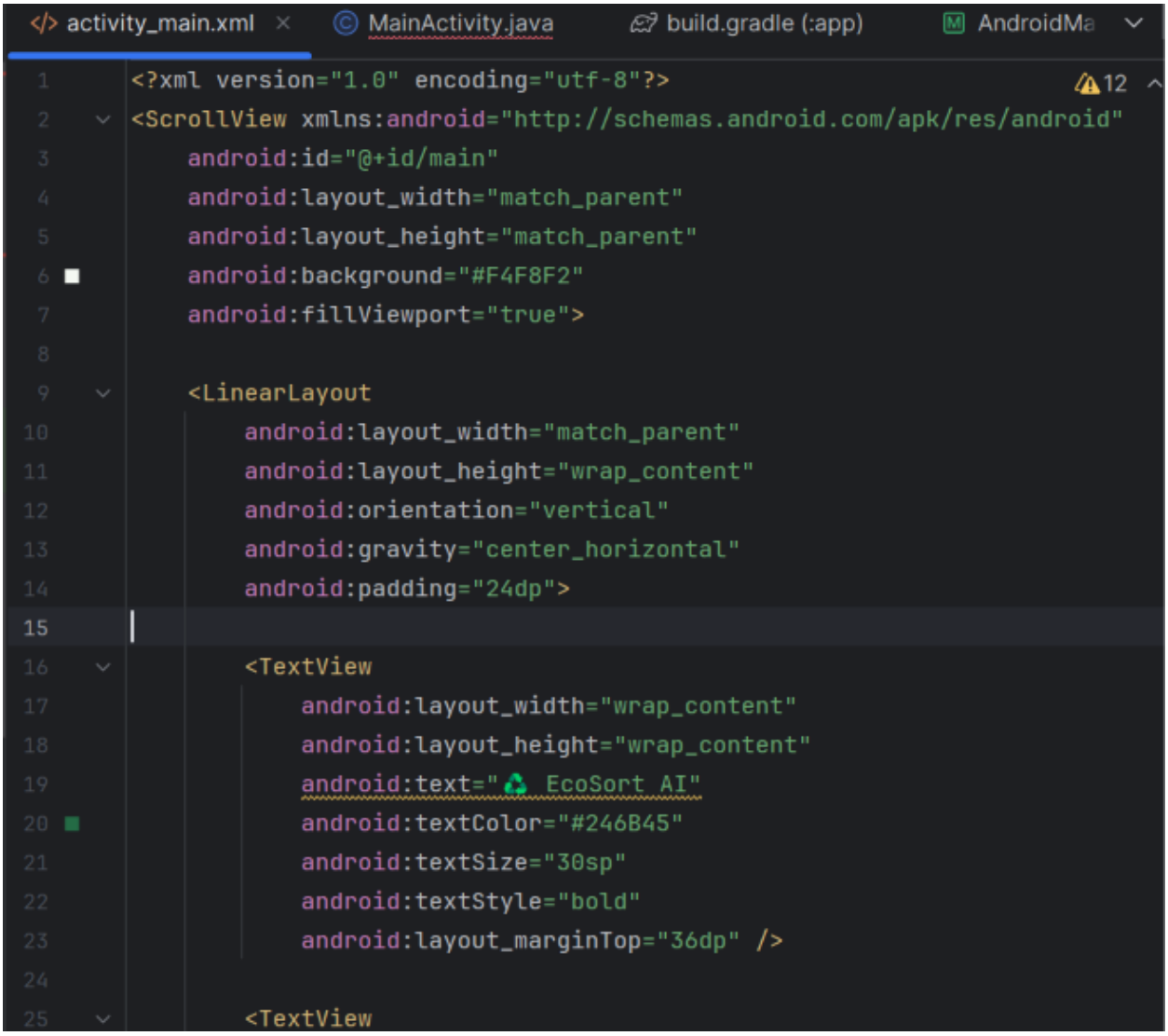
```
    }

    String predictedLabel = labelList.get(maxIndex);
    int confidence = (int) (maxProbability * 100);
    resultText.setText("Prediction: " + predictedLabel + "\nConfidence: " + confidenc
}

1 usage
private ByteBuffer convertBitmapToByteBuffer(Bitmap bitmap) {
    ByteBuffer byteBuffer = ByteBuffer.allocateDirect( capacity: 4 * IMAGE_SIZE * IMAGE_
    byteBuffer.order(ByteOrder.nativeOrder());
    int[] intValues = new int[IMAGE_SIZE * IMAGE_SIZE];
    bitmap.getPixels(intValues, offset: 0, bitmap.getWidth(), x: 0, y: 0, bitmap.getWi

    int pixel = 0;
    for (int i = 0; i < IMAGE_SIZE; i++) {
        for (int j = 0; j < IMAGE_SIZE; j++) {
            int val = intValues[pixel++];
            byteBuffer.putFloat( value: ((val >> 16) & 0xFF) / 255.0f);
            byteBuffer.putFloat( value: ((val >> 8) & 0xFF) / 255.0f);
            byteBuffer.putFloat( value: (val & 0xFF) / 255.0f);
        }
```

```
            byteBuffer.putFloat(
        }
    }
    return byteBuffer;
}
```


activity_main.xml
MainActivity.java
build.gradle (:app)
AndroidMa
12
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:background="#F4F8F2"
android:fillViewport="true">
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="vertical"
android:gravity="center_horizontal"
android:padding="24dp">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="♻ EcoSort AI"
android:textColor="#246B45"
android:textSize="30sp"
android:textStyle="bold"
android:layout_marginTop="36dp" />
<TextView

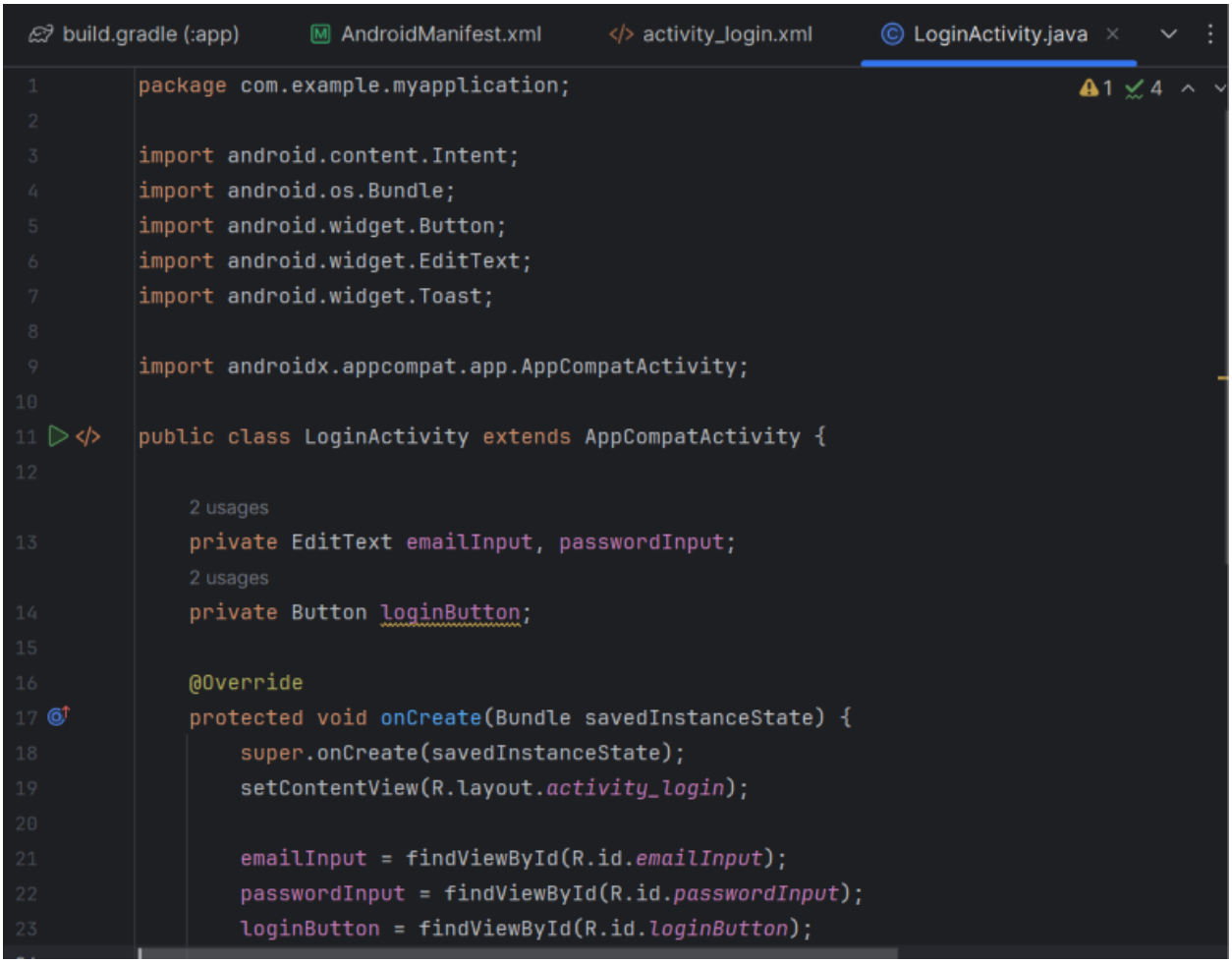
```
2     <ScrollView xmlns:android="http://schemas.android.com/apk/res/andr 12
9         <LinearLayout
24
25            <TextView
26                android:layout_width="match_parent"
27                android:layout_height="wrap_content"
28                android:text="Smart Waste Classification"
29                android:textColor="#26382D"
30                android:textSize="21sp"
31                android:textStyle="bold"
32                android:gravity="center"
33                android:layout_marginTop="28dp" />
34
35            <TextView
36                android:layout_width="match_parent"
37                android:layout_height="wrap_content"
38                android:text="Select or capture a waste image and let on-device
39                android:textColor="#607064"
40                android:textSize="16sp"
41                android:gravity="center"
42                android:layout_marginTop="12dp" />
43
44            <ImageView
45                android:id="@+id/previewImage"
46                android:layout_width="match_parent"
```

```
            android:background="#E2EBDD"
            android:text="Your AI prediction will appear here."
            android:textColor="#26382D"
            android:textSize="17sp"
            android:gravity="center" />

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="16dp"
            android:text="Categories depend on the trained model."
            android:textColor="#607064"
            android:textSize="13sp"
            android:gravity="center" />

    </LinearLayout>
</ScrollView>
```

LoginActivity configuration:


build.gradle (:app)
AndroidManifest.xml
activity_login.xml
LoginActivity.java
1
4
package com.example.myapplication;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
public class LoginActivity extends AppCompatActivity {
2 usages
private EditText emailInput, passwordInput;
2 usages
private Button loginButton;
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_login);
emailInput = findViewById(R.id.emailInput);
passwordInput = findViewById(R.id.passwordInput);
loginButton = findViewById(R.id.loginButton);

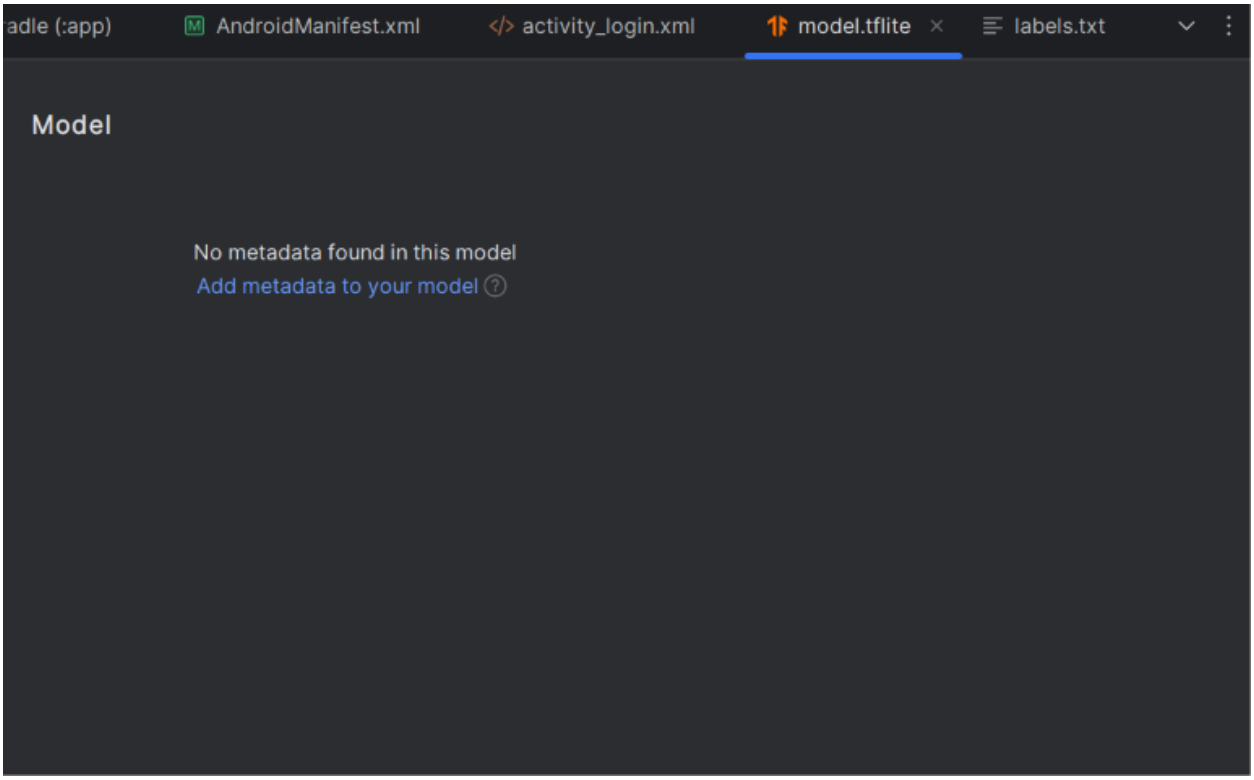
```
20
21          emailInput = findViewById(R.id.emailInput);
22          passwordInput = findViewById(R.id.passwordInput);
23          loginButton = findViewById(R.id.loginButton);
24
25          loginButton.setOnClickListener( View v -> {
26              String email = emailInput.getText().toString().trim();
27              String password = passwordInput.getText().toString().trim();
28
29              if (email.isEmpty() || password.isEmpty()) {
30                  Toast.makeText( context: LoginActivity.this,  text: "Please fill all fie
31              } else {
32                  // Filhal direct MainActivity par bhej rahe hain (Next step mein Fir
33                  Toast.makeText( context: LoginActivity.this,  text: "Login Successful!",
34                  Intent intent = new Intent( packageContext: LoginActivity.this, MainAct
35                  startActivity(intent);
36                  finish();
37              }
38          });
39      }
40  }
```

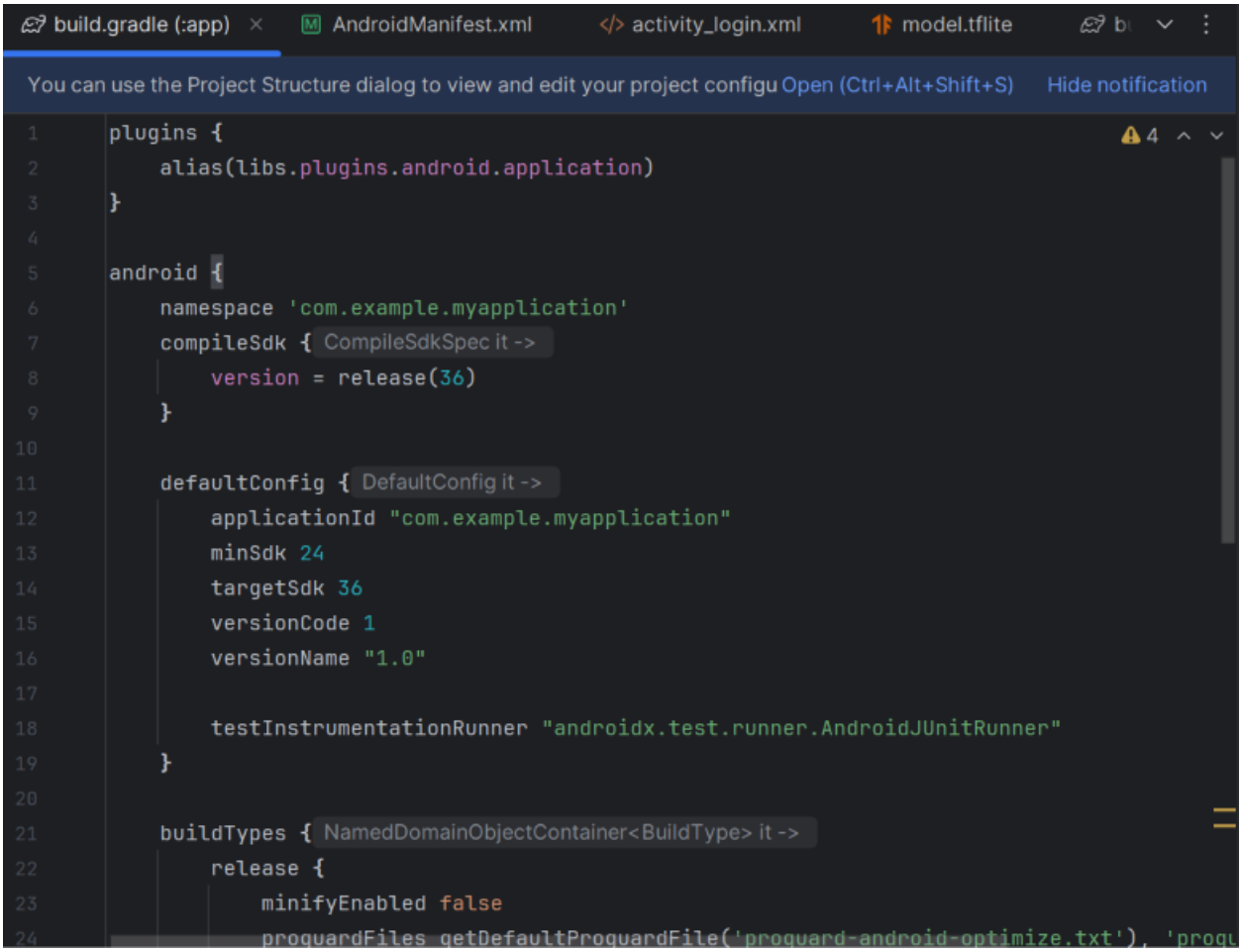
```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#101426"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="24dp">

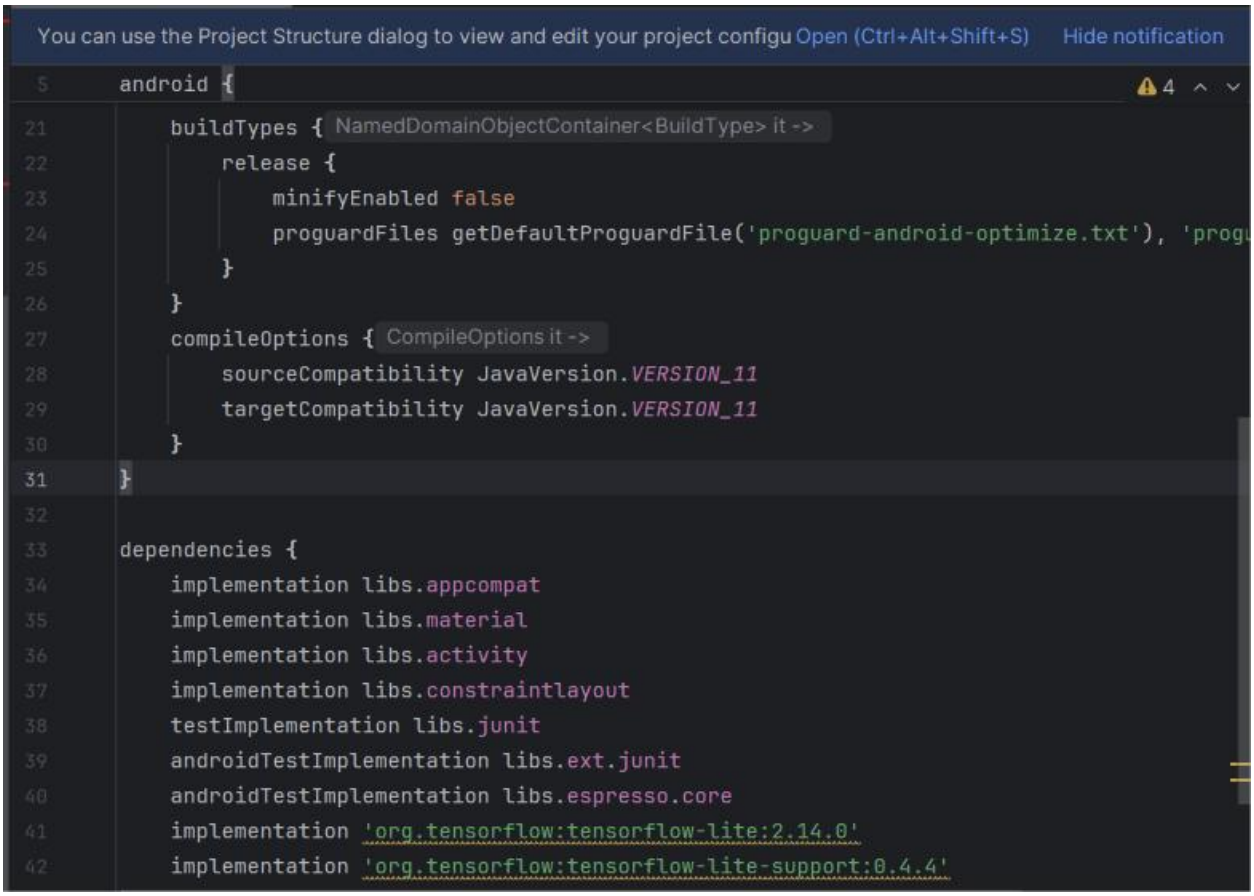
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome Back!"
        android:textColor="#FFFFFF"
        android:textSize="26sp"
        android:textStyle="bold" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp"
        android:layout_marginBottom="32dp"
        android:text="Login to EcoSort AI"
        android:textColor="#A0AEC0"
```

TensorFlow Lite Model Loading:


adle (:app)
AndroidManifest.xml
activity_login.xml
model.tflite
labels.txt
Model
No metadata found in this model
Add metadata to your model


build.gradle (:app)
AndroidManifest.xml
activity_login.xml
model.tflite
You can use the Project Structure dialog to view and edit your project configu Open (Ctrl+Alt+Shift+S) Hide notification
plugins {
alias(libs.plugins.android.application)
}
android {
namespace 'com.example.myapplication'
compileSdk { CompileSdkSpec it ->
version = release(36)
}
defaultConfig { DefaultConfig it ->
applicationId "com.example.myapplication"
minSdk 24
targetSdk 36
versionCode 1
versionName "1.0"
testInstrumentationRunner "androidx.test.runner.AndroidJUnitRunner"
}
buildTypes { NamedDomainObjectContainer<BuildType> it ->
release {
minifyEnabled false



Conclusion

The **Number Guessing App** demonstrates the integration of a Deep Learning model into an Android application using **TensorFlow Lite**. The key components of the application include the Gradle configuration, user authentication, canvas drawing functionality, and TensorFlow Lite model loading.

This project provides practical hands-on experience in implementing handwritten digit recognition through on-device machine learning. Furthermore, it showcases how native Android development and TensorFlow Lite can be combined to build an interactive, AI-powered mobile application.